

Basic Server Administration

Module 8.4

- 8.4.1 Service Configuration Files
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- 8.4.3 Monitoring Service Logs

Service Configuration Files

- 1 Linux service config files live in `/etc/` — e.g., `/etc/ssh/sshd_config`, `/etc/nginx/nginx.conf`
- 2 Config files use plain-text syntax: key-value pairs, block directives, or INI-style sections
- 3 After editing, reload or restart the service: `systemctl restart <service-name>`
- 4 Always validate syntax before applying — `nginx -t`, `sshd -T`, `apachectl configtest`
- 5 Back up configs before changes; consider `etckeeper` or `git` to track `/etc/` history

Hardening Devices

- 1 Disable unneeded services to reduce the attack surface: `systemctl disable <service>`
- 2 Enforce a firewall policy — allow only required ports using `ufw`, `firewalld`, or `iptables`
- 3 Harden SSH: set `PermitRootLogin no` and enforce public-key authentication in `sshd_config`
- 4 Apply the principle of least privilege — run each daemon under a dedicated non-root account
- 5 Keep software current: `apt update` && `apt upgrade`; enable automatic security updates

Monitoring Service Logs

- 1 System logs are written to `/var/log/` — key files: `syslog`, `auth.log`, `kern.log`, `messages`
- 2 `journalctl -u <service>` displays logs for a specific systemd unit; `-f` follows in real time
- 3 `tail -f /var/log/syslog` streams new entries live; `grep` filters by keyword or severity
- 4 Log rotation is managed by `logrotate`; configure retention in `/etc/logrotate.d/`
- 5 `rsyslog` and `syslog-ng` support centralised logging, forwarding events to a remote server